React(JS) Clock

Presented by

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Our Goals

- Make an app with all necessary time tools in one place
- Make setting alarms or timers easy with a familiar environment
- Create UI that is nice to use and look at

Success

An app where users can easily and efficiently manage their time using and navigating our application.

Typical Users

- Students
  - They need to manage studying time and breaks.
  - They use the app by setting a study timer/alarm.
- Teachers
  - They may need to manage class activities time
  - They may use the app to set timers/alarms for students
- Office workers
  - They may need to set an alarm for a meeting or a timer for breaks
  - They may use the app to manage time at work

Development Environment

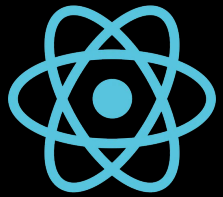
Node.js v24.22

Runs Javascript locally



ReactJS v19.3

Library used to build views



Vite v8.3

Runs app locally



NPM v11.19

Package manager



Features

The app can:

- Display a clock
- Show AM and PM
- Show the date
- Start/Stop/Repeat timer
- Set/Delete alarm
- Send notifications
- Play sound when done
- Settings to allow customization
- Snoozing alarm
- Save past set timers

Priorities

This is very similar to our feature list. Our priorities included:

- Having a working clock
- Being able to set a timer, stop it, and cancel it
- Being able to set an alarm and cancel it
- Have some sort of notification ex: pop up or play a sound

App views & Design

- ★ We based our design off of Apple's UI and Clock.

- ★ We have four views and they are as followed:
  - Clock
    - Shows the current local time
  - Alarm
    - Set alarm/Cancel alarm
    - Snooze/Stop alarm
  - Timer
    - Set Timer/Cancel Timer
    - Stop/Repeat
    - Based off of Apple's design
  - Settings
    - Change color of clock
    - add text shadow
    - time format
    - show/remove seconds
    - pick & edit background

CRUD

- Create
  - Add new alarm
- Read
  - Display saved alarm
- Update
  - Snooze, Change time
- Delete
  - Remove Alarm

Validation

Some of the validations include:

- Required valid time inputs
- Prevent a 0 minute/second timer

Challenges Faced

Carlos

- Web app would not run
  - I set the folder up incorrectly
  - I also tried to run it from the wrong directory
- I'm still new to git so when I first tried to commit it would commit to my own profile instead of our shared project
- Screen would turn black on errors had to find errors
  - Was syntax or server was not responding
- Vite server would not update
  - Had to restart server
- Alarm sound would not play
  - Had to go through code and check where error happened.
  - syntax error

Julio

- New to React
- Installing and getting the correct version of dependencies
- Getting project to run on computer
- Moving file folders caused files to delete after pushing to repo
- Getting countdown circle to smoothly animate
- The timer not being accurate while counting down